

One Dimensional Internal Ballistics

The aim of this coursework is to evaluate the internal ballistics of a solid rocket motor with a cylindrical tubular grain by applying a one-dimensional model. So, making the assumptions listed in the text one can simulate a mock-up version of P80 grain configuration, since it is an aft-finocyl but the analysis is made on a cylindrical tubular combustion chamber like the one in fig.1.

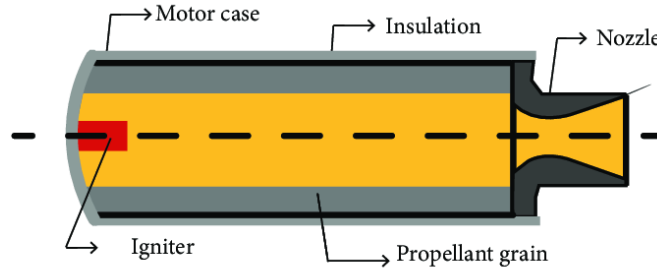


Figure 1: Tubular grain geometry.

Step 1: Variation of properties along the grain and nozzle length

Initial grain shape

From the knowledge of the grain characteristics (initial chamber volume, $V_{c_{in}}$, and propellant volume, V_p), the motor case diameter, D_{case} , and the conical nozzle characteristics (throat diameter, D_t , and half-cone angles for both the convergent and divergent part, α_{conv} and α_{div}), the initial port area A_p and the grain length L_p can be computed assuming a cylindrical combustion chamber. So, one can evaluate them from the solution of a system of two equations:

$$\begin{cases} V_c = A_p L_p + V_{conv} = 8.68 \text{ m}^3 \\ V_p = (A_{case} - A_p) L_p = m_p / \rho_p = 49.16 \text{ m}^3 \end{cases}$$

where

$$\begin{aligned} V_{conv} &= \frac{\pi L_{conv}}{3} (R_{case}^2 + R_{case} R_t + R_t^2) = 3.52 \text{ m}^3 \\ L_{conv} &= |x_a - x_t| = \frac{R_{case} - R_t}{\tan(\alpha_{conv})} = 1.25 \text{ m} \\ L_{div} &= |x_a - x_e| = \frac{R_e - R_t}{\tan(\alpha_{div})} = 2.78 \text{ m} \end{aligned}$$

where V_{conv} is the volume of the convergent part of the nozzle, L_{conv} is its length and L_{div} is the length of the divergent part. So, from the previous system one obtain that:

$$\begin{cases} L_p = \frac{V_p + V_c + V_{conv}}{A_{case}} = 7.68 \text{ m} \\ A_p = \frac{V_c - V_{conv}}{L_p} = 0.663 \text{ m}^2 \end{cases}$$

Finally, one can evaluate the so-called initial throat-to-port area ratio, J ;

$$J = \frac{A_t}{A_p} = 0.291$$

Thanks to this fundamental parameter, the Mach number at the end section of the tubular grain, M_a , where a comes from the definition of this particular section called aft-end, can be evaluated and from this also the total pressure losses along the grain. Furthermore, once $p_0(x_a)$ is known one can evaluate also the head-end pressure p_h . These relations can be seen from the definition of J as a function of M_a and M_t :

$$J = \frac{f(M_a)}{f(M_t)} \quad \text{where} \quad f(M) = \frac{M}{(1 + \delta M^2)^{\frac{\gamma+1}{2(\gamma-1)}}} \quad (1)$$

And assuming that the nozzle is choked. i.e. with $M_t=1$, one can obtain from eq.1 that $M_a=0.177$. Consequently, the head-end pressure is determined from:

$$\frac{p_0(x)}{p_h} = \frac{[1 + \delta M^2(x)]^{\frac{\gamma}{\gamma-1}}}{1 + \gamma M^2(x)} \longrightarrow p_h = p_c \frac{1 + \gamma M_a^2}{(1 + \delta M_a^2)^{\frac{\gamma}{\gamma-1}}} \quad (2)$$

which yields to $p_h = 20.31 \text{ bar}$. The chamber pressure appearing in the above equation, that is the total pressure at the grain aft-end, is computed using the equilibrium chamber pressure definition, obtained from the zero-dimensional steady-state model:

$$p_c = p_{c,eq} = (\rho_p a c^* K)^{\frac{1}{1-n}} = 19.97 \text{ bar}$$

where inside the klemmung is hidden the value of the burning surface S_b at the beginning of operations, defined as:

$$S_b = l_b L_p = 2\pi R_p L_p = 22.16 \text{ m}^2$$

where l_b is the port area perimeter.

In order to evaluate the variation of the various parameters that are needed, one must talk about the model used to describe the flow inside the port region and the nozzle.

The closed form equations for the **flow in the grain**, modelled as a flow with only mass addition (in particular with the total temperature of the injected mass equal to T_f and its velocity

perpendicular to the direction of the flow), can be written as follows:

$$\begin{aligned}\rho u &= cx \\ \rho u^2 + p &= p_h = \text{const.} \\ h + \frac{u^2}{2} &= h_f = c_p T_f = \text{const.}\end{aligned}$$

where

$$c = \frac{\rho_p r l_b}{A_p} \quad \text{and} \quad r = a p_{c,eq}^n = 6.12 \text{ mm/s}$$

with the burning rate considered as constant and calculated using the equilibrium chamber pressure. Since the flow with only mass addition is non-isentropic, a loss in the total pressure is experienced along x . At the end, using the state equation for ideal gas and the isentropic relations, one can combine the governing equations and obtain all the quantities as a function of x :

$$\begin{aligned}p(x) &= \frac{p_h}{\gamma + 1} \left\{ 1 + \sqrt{1 + (\gamma^2 - 1) \left[1 - 2h_f \left(\frac{cx}{p_h} \right)^2 \right]} \right\} \\ M(x) &= \left[\frac{1}{\gamma} \left(\frac{p_h}{p(x)} - 1 \right) \right]^{\frac{1}{2}} \\ p_0(x) &= p(x) [1 + \delta M(x)^2]^{\frac{\gamma}{\gamma-1}} \\ T(x) &= \frac{T_f}{1 + \delta M(x)^2} \\ G(x) &= \rho(x) u(x) = cx \\ u(x) &= \frac{G(x) RT(x)}{p(x)} = M(x) \sqrt{\gamma RT(x)} \\ r(x) &= a p(x)^n\end{aligned}$$

The closed-form equations for the **flow in the nozzle**, assuming a choked throat condition, can be obtained imposing the only area change as the source term, thus they refer to an isentropic flow, where the total pressure is conserved

$$\begin{aligned}\frac{A(x)}{A_t} &= \frac{1}{M(x)} \left[\frac{1 + \delta M(x)^2}{1 + \delta} \right]^{\frac{\gamma+1}{2(\gamma-1)}} \\ p_c &= p(x) [1 + \delta M(x)^2]^{\frac{\gamma}{\gamma-1}} = p_0(x_a) = \text{const} \\ T_c &= T(x) [1 + \delta M(x)^2] = T_f = \text{const}\end{aligned}$$

From these equations one can evaluate $M(x)$ from the area law and substituting this value of the Mach number to obtain the profiles of $p(x)$, $T(x)$, $u(x)$, $G(x)$.

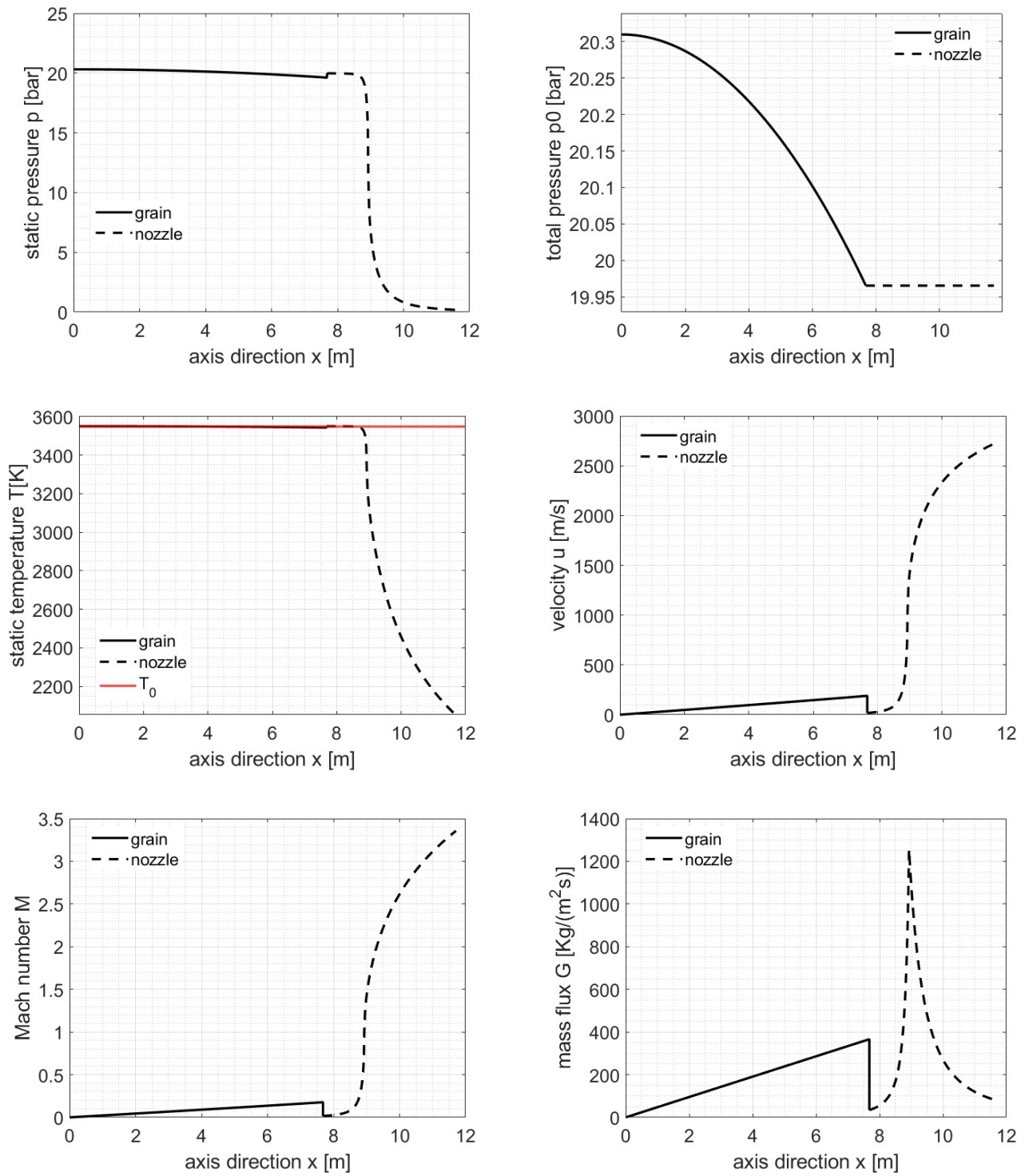


Figure 2: Initial grain configuration profiles along the motor axial direction.

Final grain shape

The results are obtained with the same procedure of the initial grain configuration but in this case all the propellant has been burned. So, the calculations are done imposing the same L_p and V_c but now, since there is no more propellant, $V_p=0$ and $R_p \rightarrow R_{case}$.

Comparison

The results are shown in fig.2 for the initial configurations and in fig.3 for the final configuration. Starting from the initial configuration, one can describe the spatial evolution of the flow along the grain, which is represented in the plots with a continuous line. In this case the static pressure undergoes a distributed decrease because of the increase in the flow velocity. Since the flow is not isentropic, there is also a total pressure drop, which stops in the aft-end section. Both static and total pressure losses are limited due to the low value of J , which reflects on a limited value of M_a and then on low total pressure losses (numerically explained in eq.2) which clearly reflects on the value of the static pressure. The temperature is almost unaffected by the flow evolution within the port cavity and remains almost equal to the adiabatic flame temperature (i.e. the total temperature T_0). The mass flux increases, as expected, in a linear way, as well as the Mach and velocity.

When the analysis of the flow inside the nozzle starts, all the quantities shows a discontinuity at the aft-end section, because this is the point where mass addition stops and the only acting source term is the one related to the area variation. In particular, up to the port area the flow is subsonic, then it encounters a sudden enlargement in terms of area which results in a sudden compression. The consequence is an immediate decrease of the velocity, Mach number and mass flux, while pressure and temperature remain unchanged. The flow in the nozzle, which is represented in the plots with a dotted line, follows the typical expansion process expected from the ideal nozzle theory. At the end the Mach number is very high, just above 3, while the static pressure is of the order of the ambient pressure. The mass flux, which is a mass flow rate per unit area, clearly has an opposite behaviour with respect to the area change: it increases until the throat area, where it reaches its maximum value, and then in the divergent part of the nozzle it decreases. This can be an issue from the point of view of thermal loads on the nozzle walls: from the boundary layer theory it is known that the heat transfer coefficient in the nozzle has its maximum where the mass flux reaches its maximum value. In order to overcome this problem, usually a larger thickness of thermal protection material is put corresponding to this critical section (ablative materials).

For the final configuration, one can notice that the discontinuity in the geometry is not present and this reflects in the absence of the jumps at the aft-end section. Moreover, since the burning surface is larger than that of the first case, the value of static pressure all along the port cavity is very large but its variation along the grain length is almost negligible, due to value of $J=0.0273$ at the end. Another characteristic is that the velocity up to the end of the grain is almost negligible with respect to the initial grain configuration because the port area is larger.

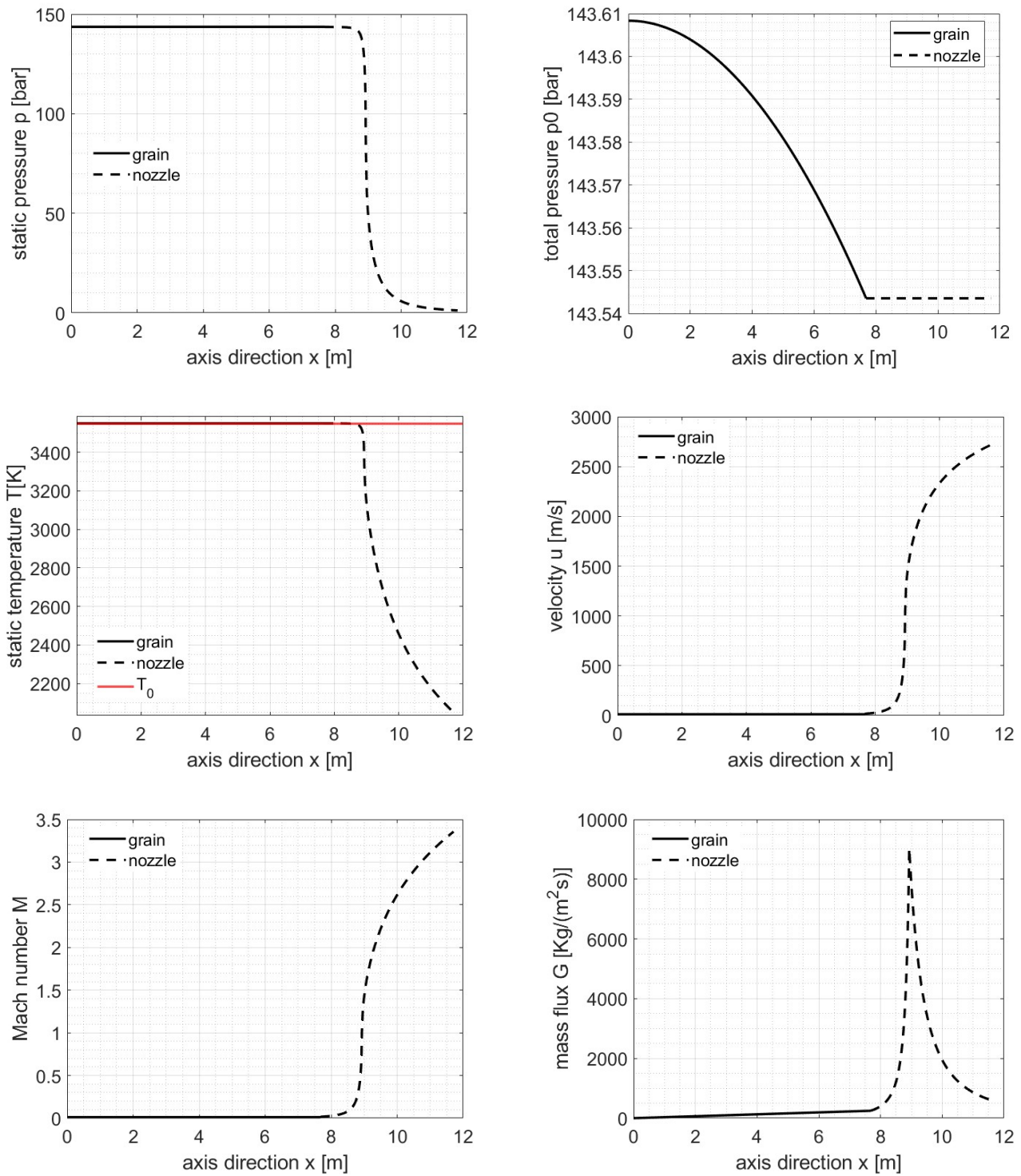


Figure 3: Final grain configuration profiles along the motor axial direction.

The comparison of the burning rate is shown in fig.4, where one can observe that, for both the initial and final grain configuration, the approximated r , i.e. computed by using p_{ceq} and considered constant up to $x = x_a$, is a valid guess because it is comparable with the mean value of the variable burning rate, defined as $r(x) = a p(x)^n$ with a decreasing trend that reflects the behaviour of the static pressure.

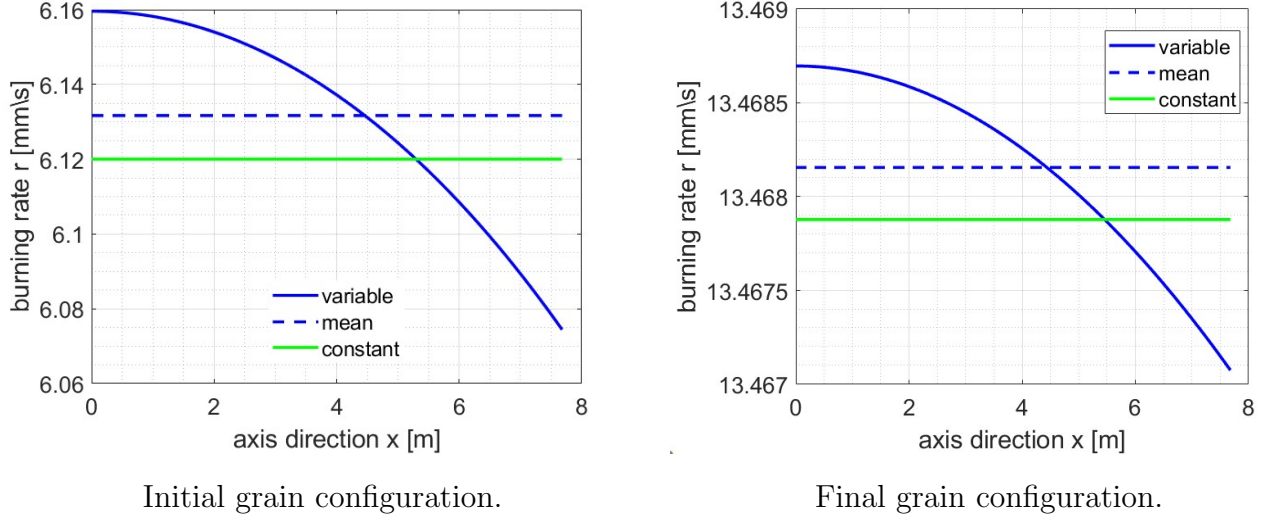


Figure 4: Burning rate profiles along x : comparison between initial and final configurations.

Step 2: Effect of increasing the volumetric loading fraction

This analysis is taken considering different geometrical constraints: V_{case} , the case volume, L_p , the propellant length, A_t , the throat area. The volumetric loading fraction is a parameter defined as:

$$V_f = \frac{V_p}{V_c + V_p} = \frac{V_p}{V_{case}}$$

so, it is interesting to evaluate how a variation of V_f affects the flow evolution along x . Since V_{case} is a constraint, in order to take into account different values of V_f one has to change the propellant volume, which is reflected in a change of the port diameter and therefore in a change of the port area A_p . All this analysis collapses in a change of the J parameter because the throat area is a fundamental constraint for thrust requirements. In the step 1 $V_f=0.85$ and the corresponding J is about 3.

For the following analysis two cases will be considered:

1. **J = 0.6** with $V_f = 0.896$;
2. The limiting case for which the port area equals the throat area, i.e. with **J = 1** and $V_f = 0.914$.

The results are shown in fig.6 for the case 1 and fig.7 for the case 2.

The general trend, which leads from the case with $J \approx 3$ of the step 1 to the cases analyzed in this step with $J = 0.6$ and $J = 1$, is driven by the reduction of the port area, which reflects in a global decrease of the values of $p(x)$ and $T(x)$ because of the effect of minor burning surface. Moreover, since the port area is lower and the velocity increases with J , as well as the Mach number, the pressure losses tend to be more significant.

The burning rate profiles are shown in fig.5. One can notice that the difference between the constant r used in the calculations and the variable one, especially at the head-end and the aft-end sections, tends to be more significant with the increase of J . This is due to the fact that the relative pressure drop along the grain is enhanced by an increase in J and this reflects in the burning rate variation along x .

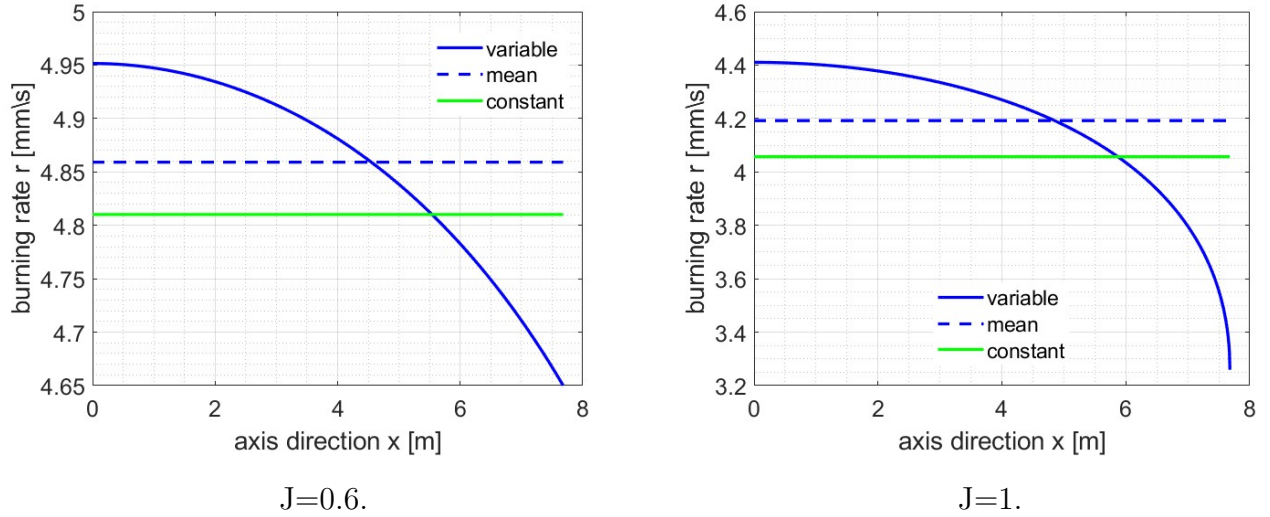


Figure 5: Burning rate profiles along x : comparison between configurations with different J .

In general, the 1D internal ballistics analysis when $J < 1$ says that the Mach at throat section M_t can be either sonic or supersonic, depending on the external pressure outside the nozzle, while the supersonic case is not admitted because the flux at the port area is necessarily subsonic and, thanks to the abrupt area change at the aft-end section and the passage in the convergent part, it can reach the sonic condition at most.

In this case:

$$J < 1 \implies A_t < A_p \implies f(M_t) < f(M_a)$$

In the limiting case where $J = 1$, $f(M_t) = f(M_a)$ and thus $M_t = M_a$. From the ideal nozzle theory this can lead to two results: if the external pressure is less than the subsonic pressure both

sections are choked, i.e. $M_a = M_t = 1$; if the external pressure is bigger than p_{sub} both M_a and M_t are equal and subsonic. But since the choked throat condition is imposed, the case $J=1$ implies $M_a = 1$.

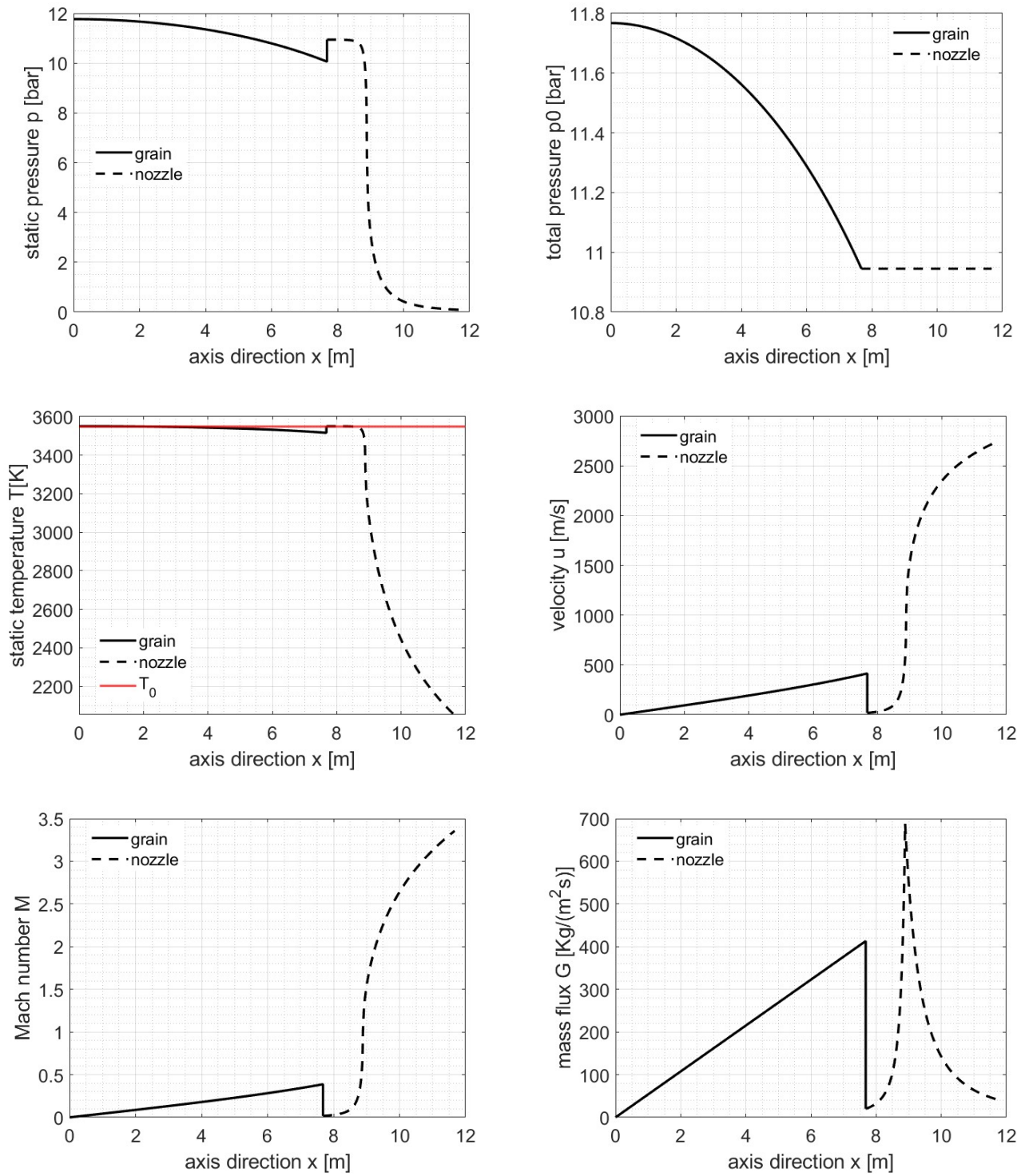


Figure 6: Profiles along the motor axial direction for the case with $J=0.6$.

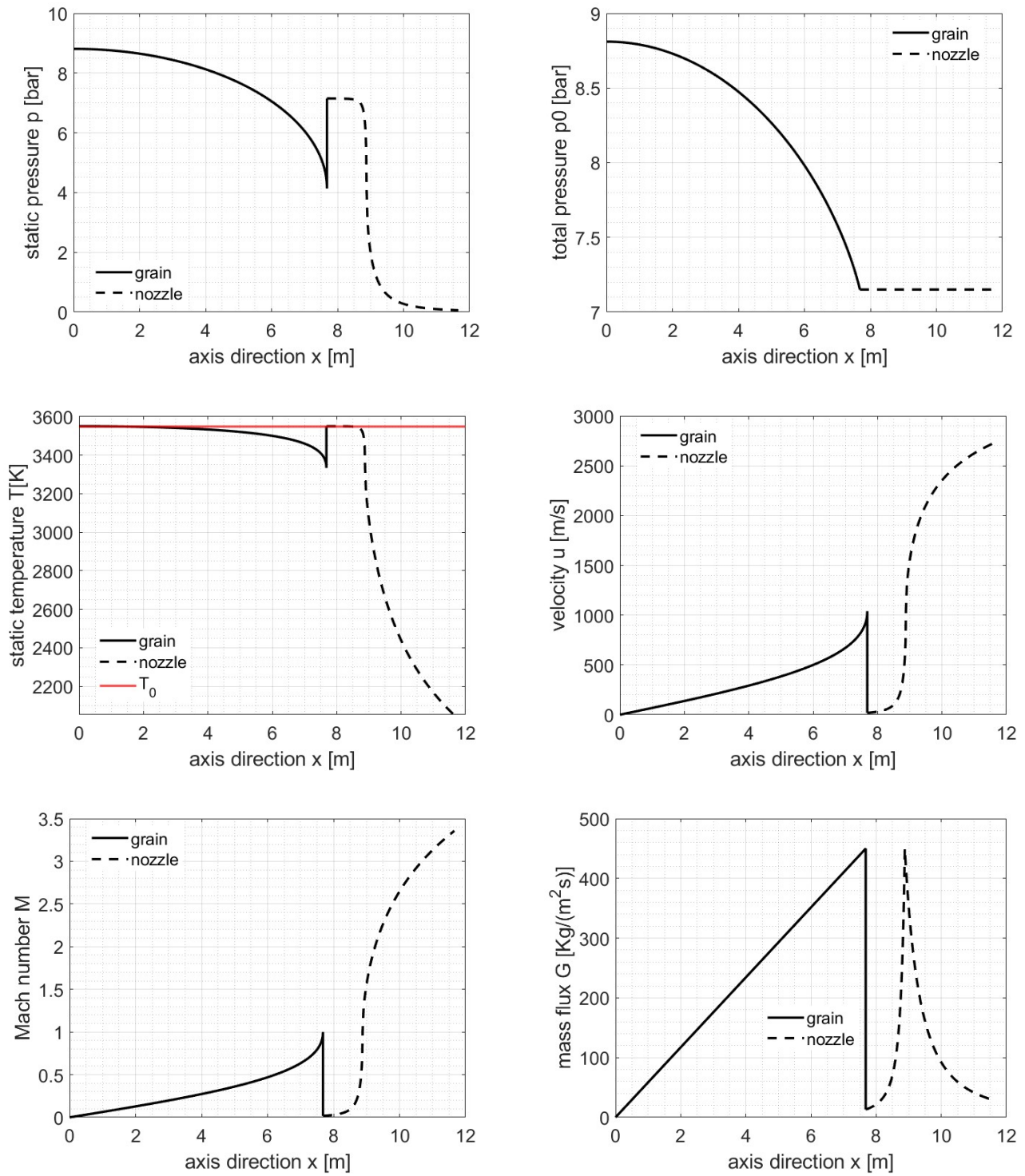


Figure 7: Profiles along the motor axial direction for the case with $J=1$.